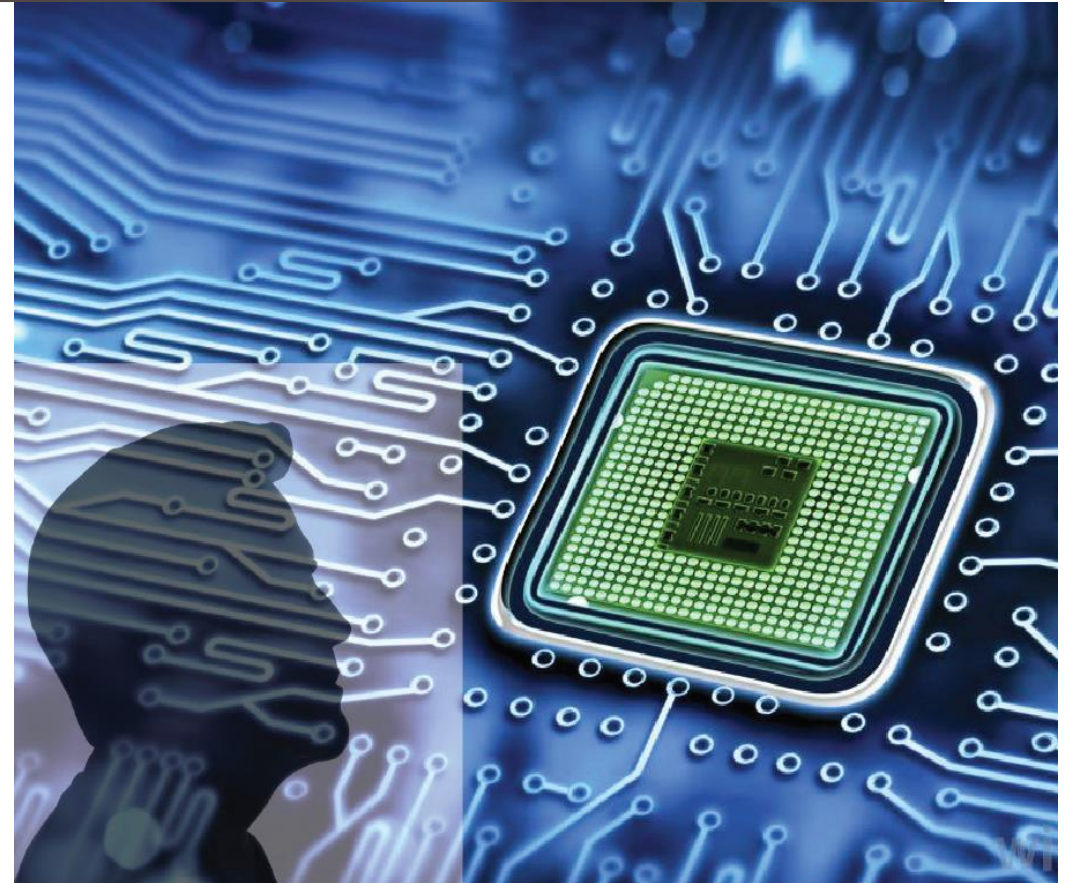


VLSI DESIGN FLOW

Week-3, Lecture-2 Introduction

Sneh Saurabh
31st August, 2024



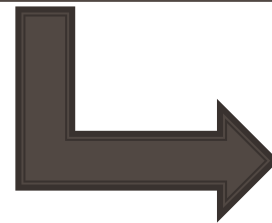
VDF: Revision/Announcements

Office Hours for TAs

- Upal - Monday - 2:00 to 3:00 PM (Library Ground Floor)
- Abhishek - Tuesday - 4:45 to 5:45 PM (R&D 6th floor)
- Anubhav - Wed - 2:00 to 3:00 PM (Library 3rd floor - Casual study)
- Sandeep - Thursday - 4:45 to 5:45 PM (Lab 617, R&D 6th floor)
- Rudraksh - Friday - 11:30 to 12:30 PM (R&D 6th floor beside stairs)

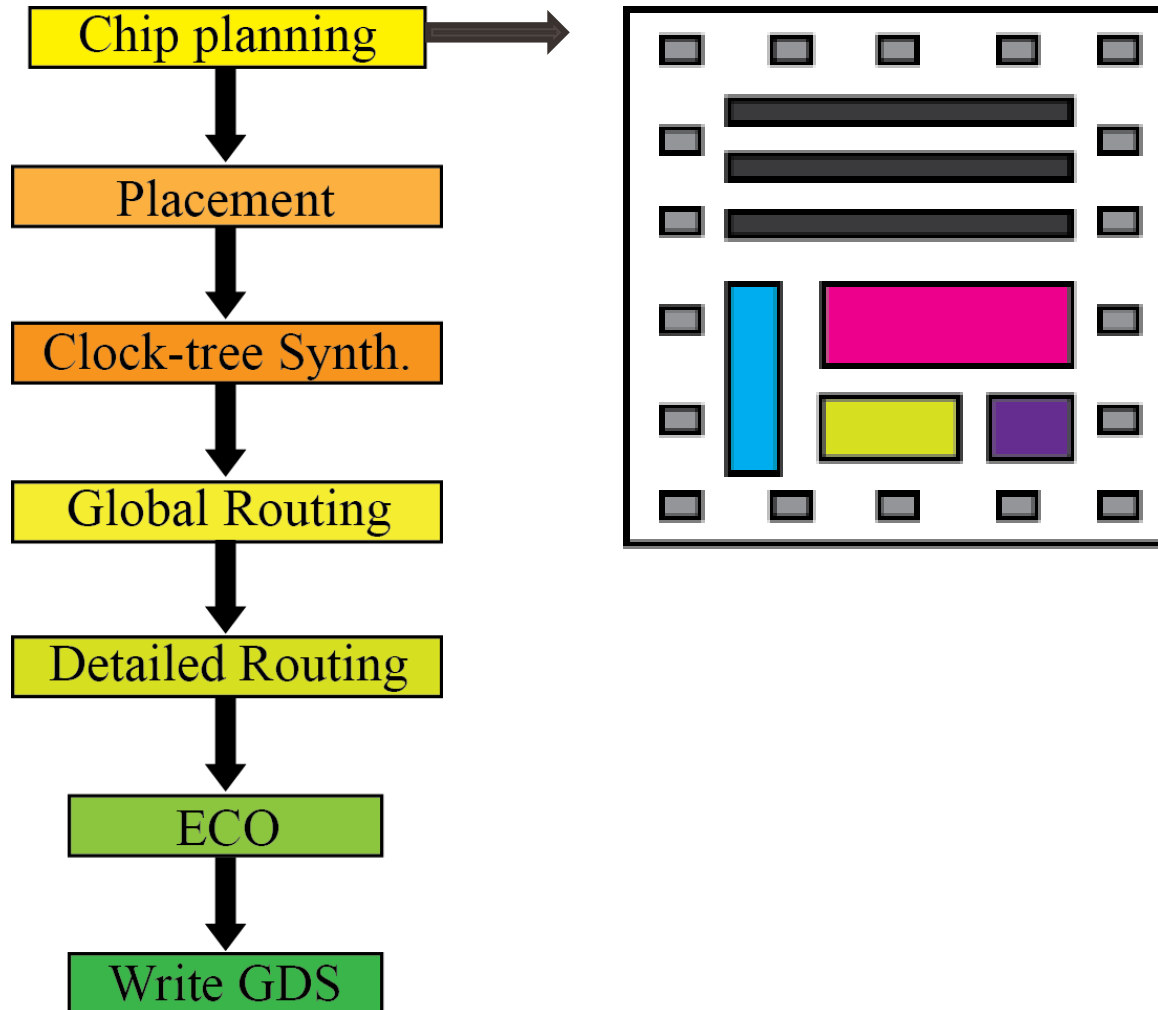


Overview of RTL to GDS Flow



Physical Design

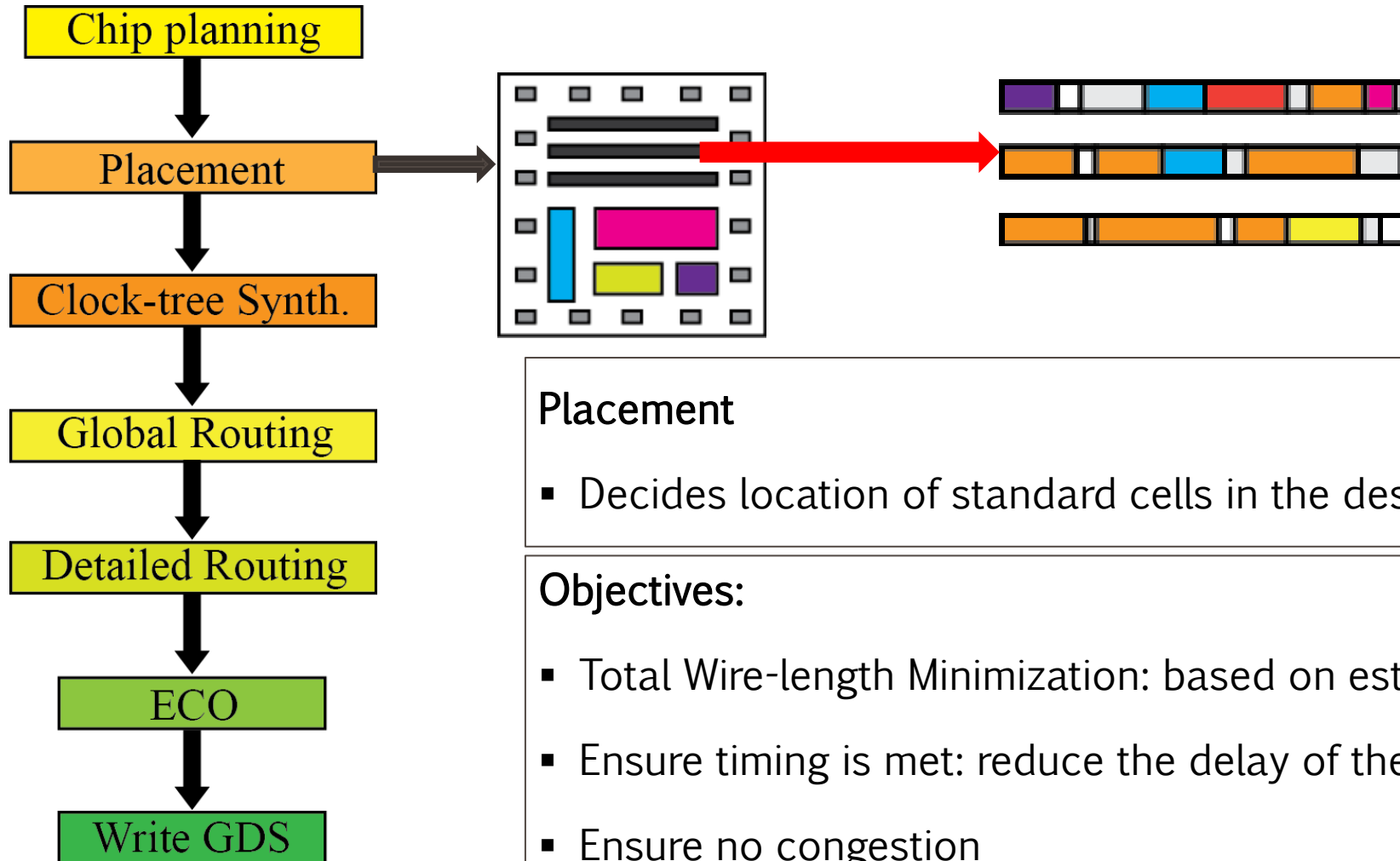
Physical Design: Chip Planning



Chip Planning

- First step in physical design in which we take major decisions
- Partitioning into subsystems or blocks
- Arrange the blocks on the chip/die
- Allocation of area for the standard cells, macros, memory
- Includes IO cell planning and Power Planning

Physical Design: Placement



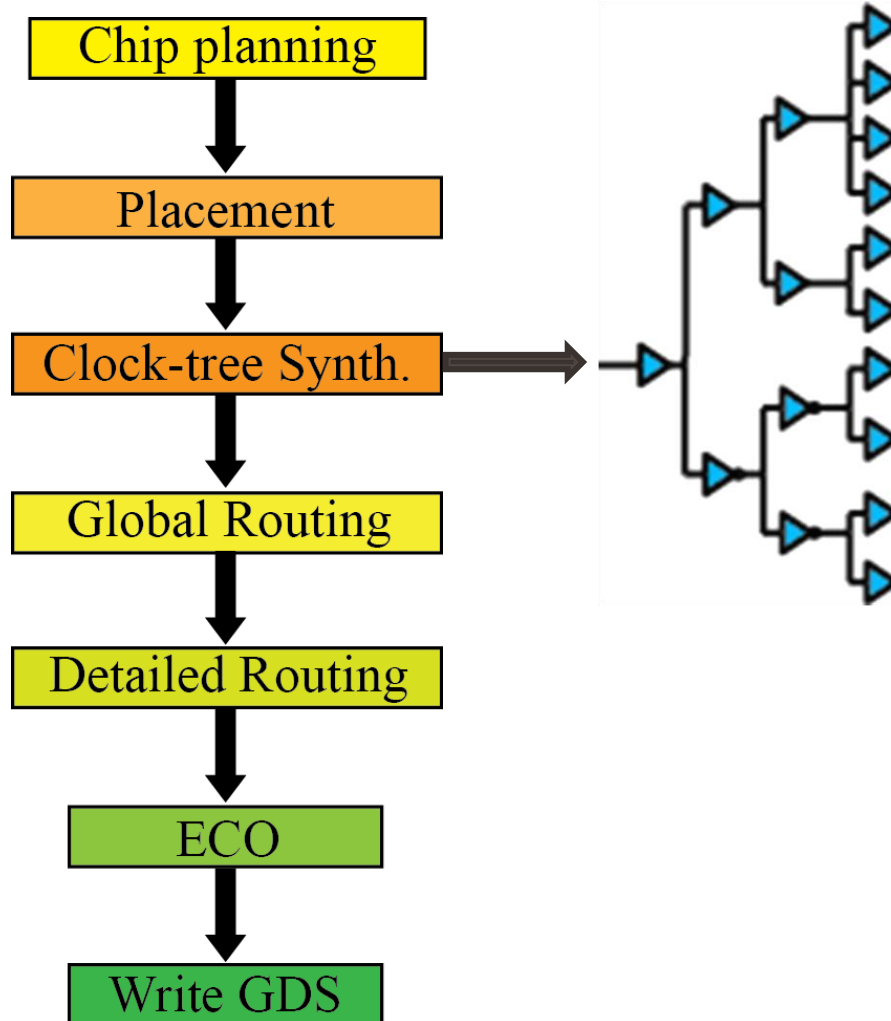
Placement

- Decides location of standard cells in the design

Objectives:

- Total Wire-length Minimization: based on estimates
- Ensure timing is met: reduce the delay of the critical path
- Ensure no congestion

Physical Design: Clock Tree Synthesis



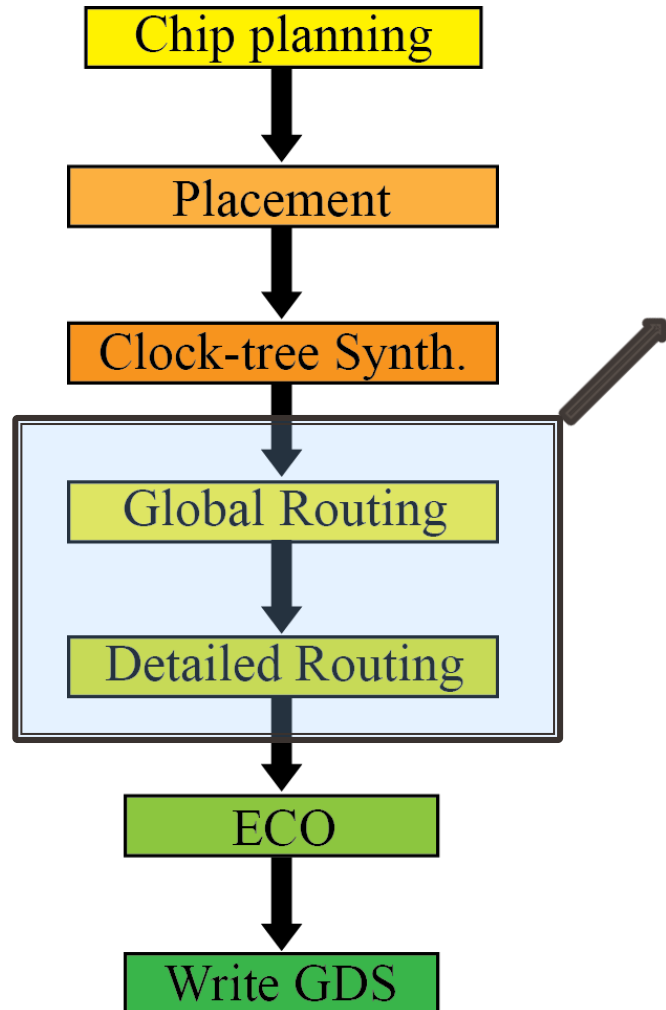
Clock Tree Synthesis (CTS)

- Decides topology of clock network and how clock reaches each clocked element
- CTS also performs wiring of clock network (avoids detour since majority of routing resources still unused)

Objectives:

- Ensure minimum skew: symmetric structure
- Minimize power dissipation: clock network consumes large fraction of total power

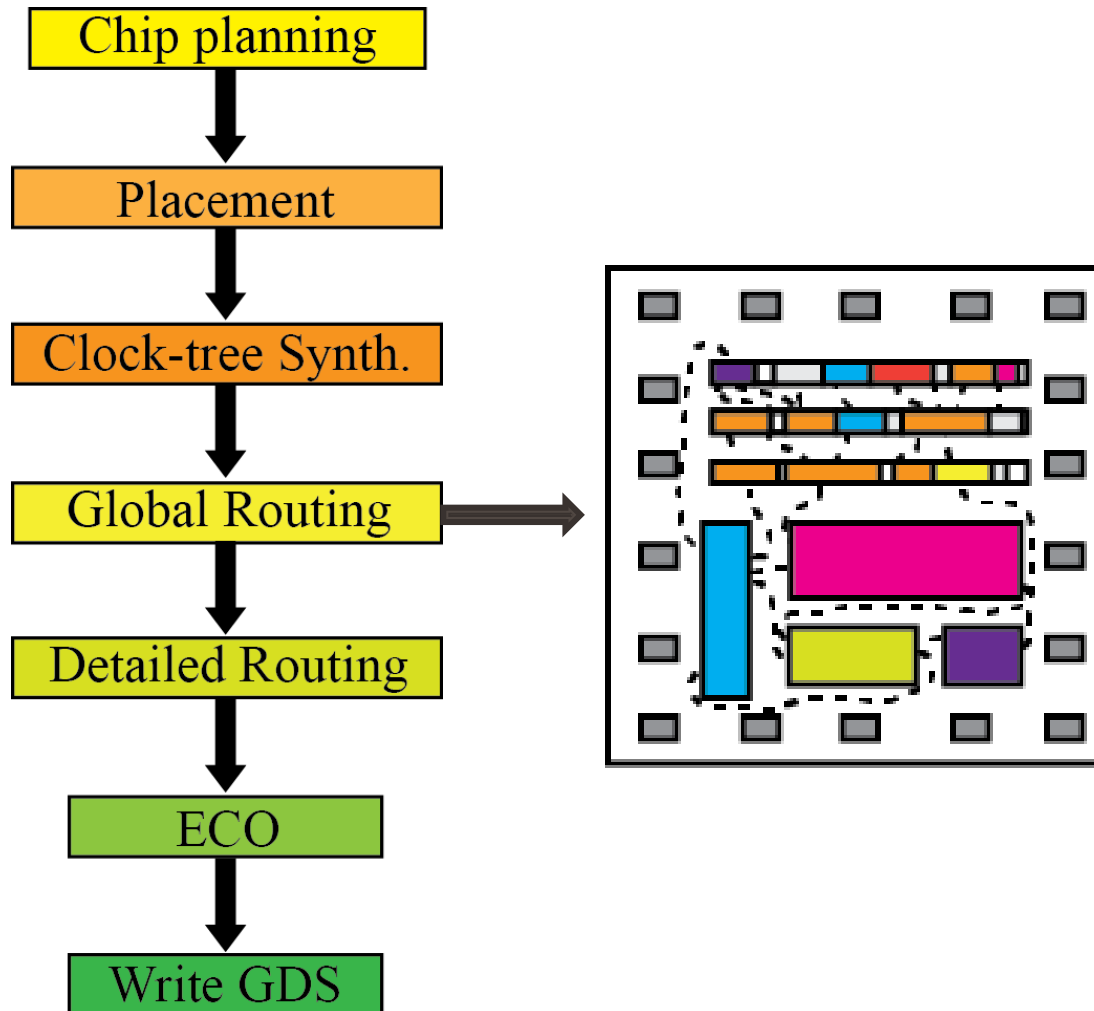
Physical Design: Routing



Routing

- Creates wire layout for all the nets (other than clock and power supply) satisfying certain constraints
 - **Objective:** Use minimum wire-length, routing area, vias
-
- Very complicated process (too many nets and routing constraints)
 - It is typically done in two steps:
 - Global Routing
 - Detailed Routing

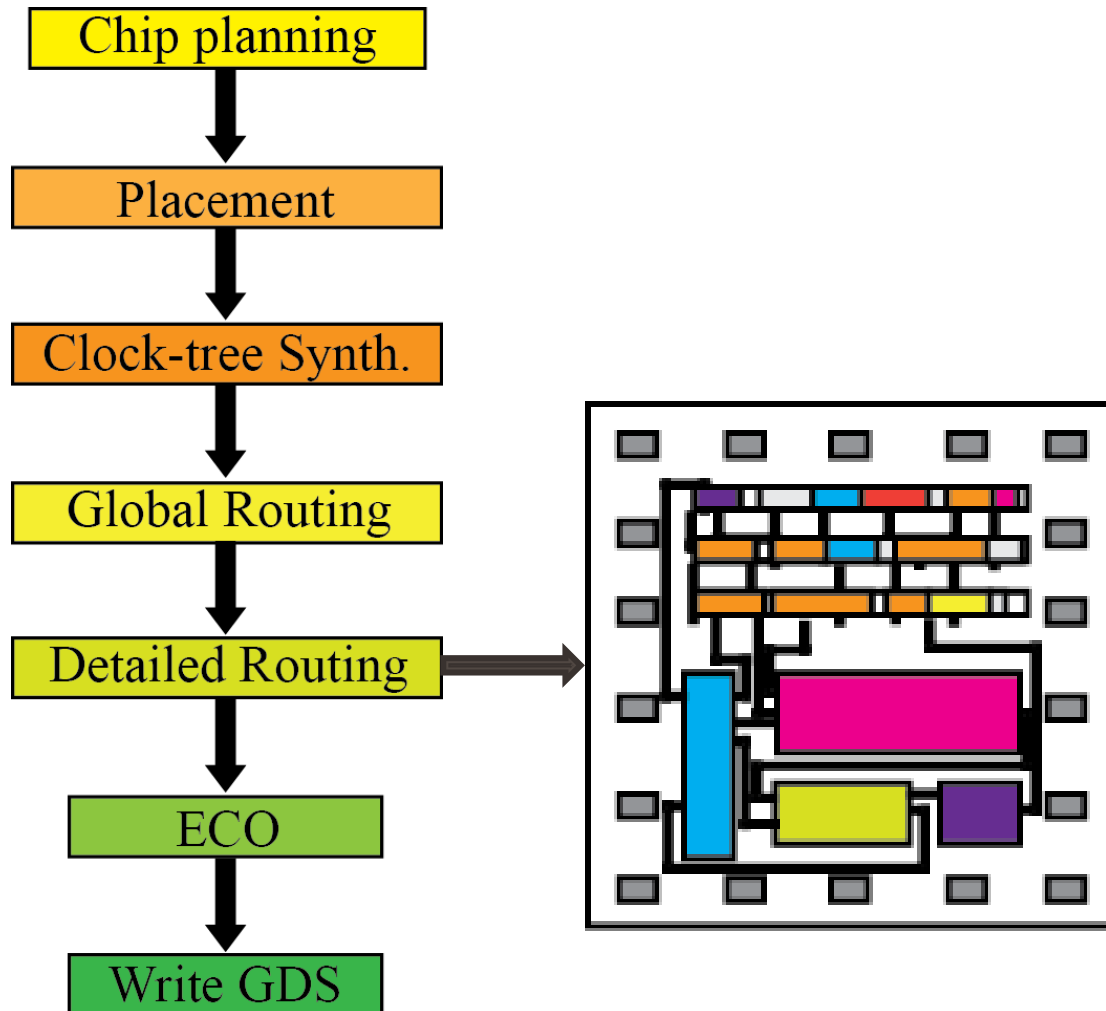
Physical Design: Global Routing



Global Routing

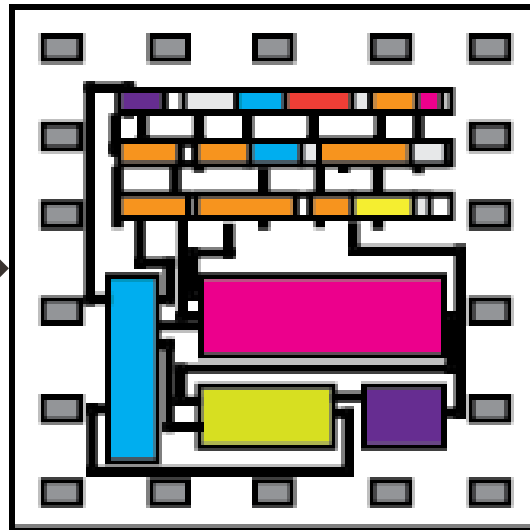
- Planning stage for routing
 - Actual layout of wires not created
 - A routing plan for a given net is created
-
- Entire routing region is partitioned into rectangular tiles or **global bins**
 - Global routing assigns a set of global bins that will be used for making connections for a given net

Physical Design: Detailed Routing

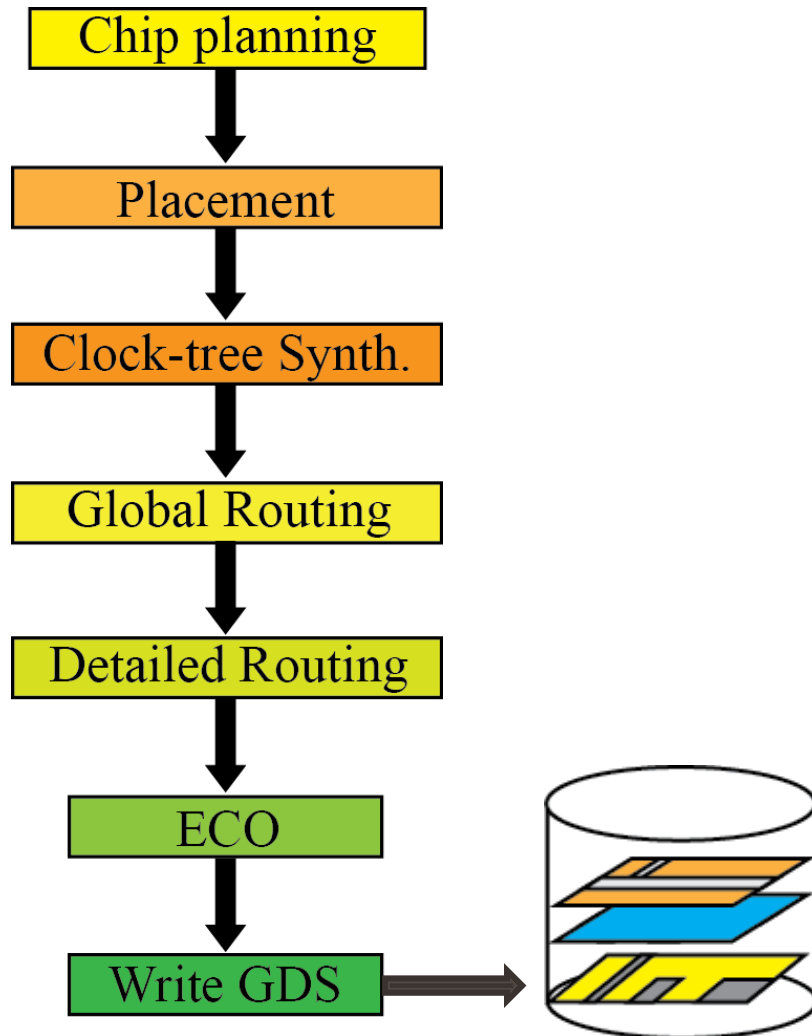


Detailed Routing

- Decides actual layout of each net in the pre-assigned global bins
- The detailed router decides the actual physical interconnections of nets by:
 - Allocating wires on each metal layer
 - Vias for switching between metal layers



Physical Design: ECO and GDS Writing



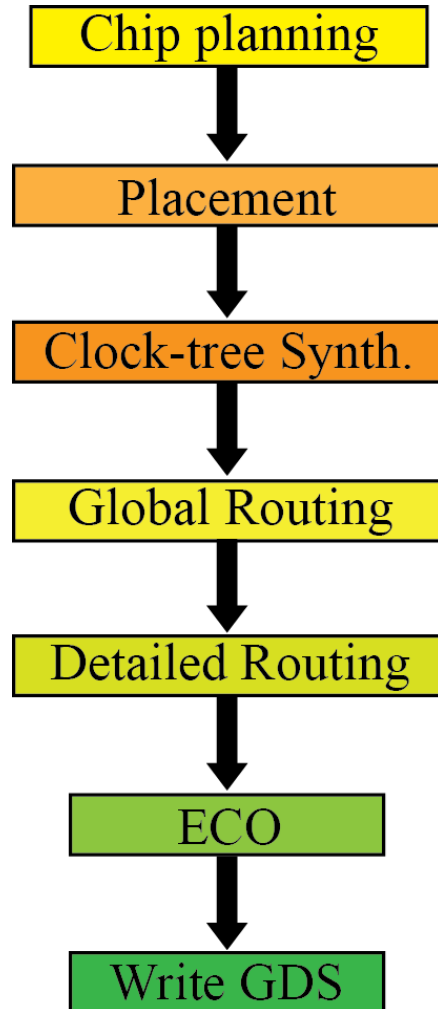
Engineering Change Order (ECO)

- Make small final fixes in the design

Write GDS

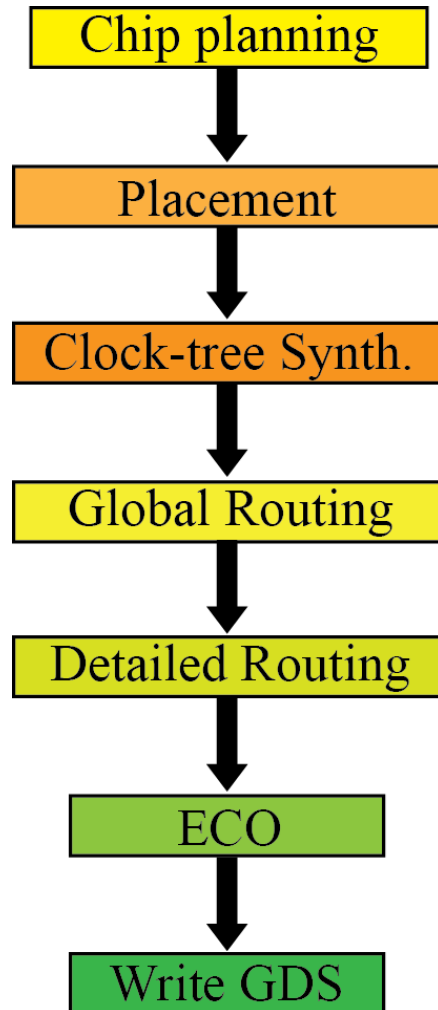
- Dump the layouts of each layer in a GDS file

Physical Design: Optimizations



- Between each physical design task there are optimization steps:
 - Small changes in the design to improve PPA
 - Example of changes in the design:
 - ✓ Buffer insertion on a given net
 - ✓ Changes in the size of a given cell
 - ✓ Changes in the placement of a given cell
 - ✓ Changes in routing for a given net

Physical Design: Iterative flow



- Physical design implementation tasks performed along with verification tasks (timing, power, signal integrity, etc.)
- Physical design implementation should ensure design closure (including *timing* closure) [all constraints are satisfied]

- Achieving design closure is challenging
 - Design tasks do not have the full information and works with estimates that can be wrong

- Physical design flow is typically **iterative**
- One task may require that previous tasks retract some design decisions
 - This creates loops in the physical design flow

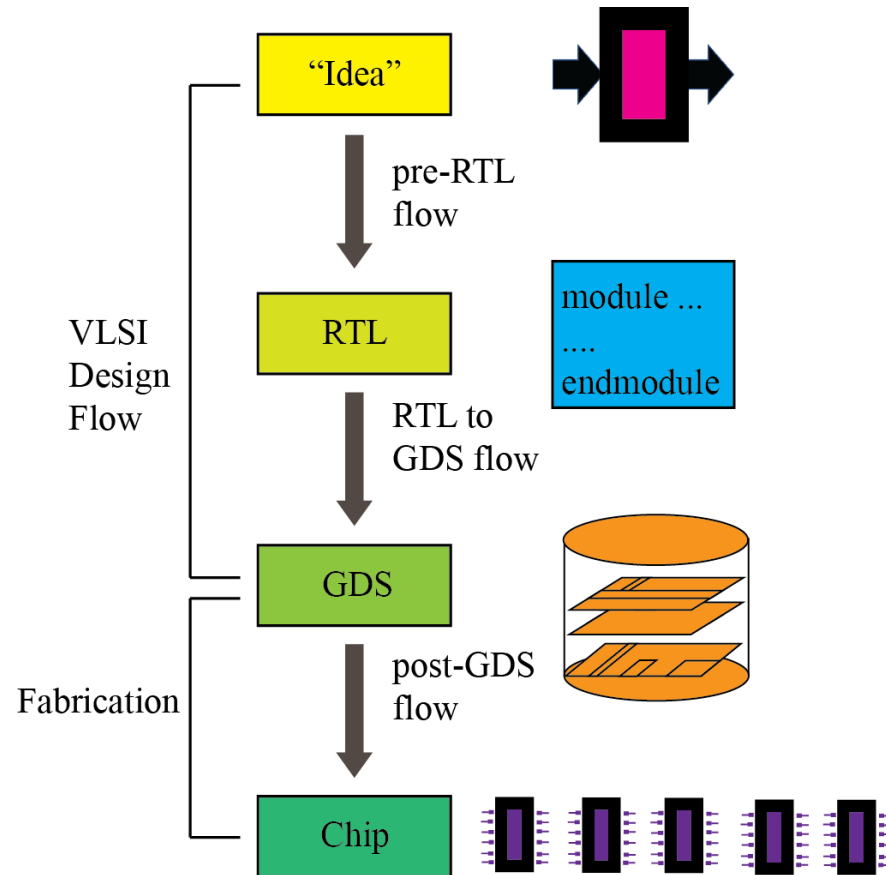
- Achieving design closure with minimum number of iterations is the goal of a physical design flow

Overview of RTL to GDS Flow



Verification

VLSI Design Flow: Verification



Verification:

We perform verification to ensure that the design works as per given functionality

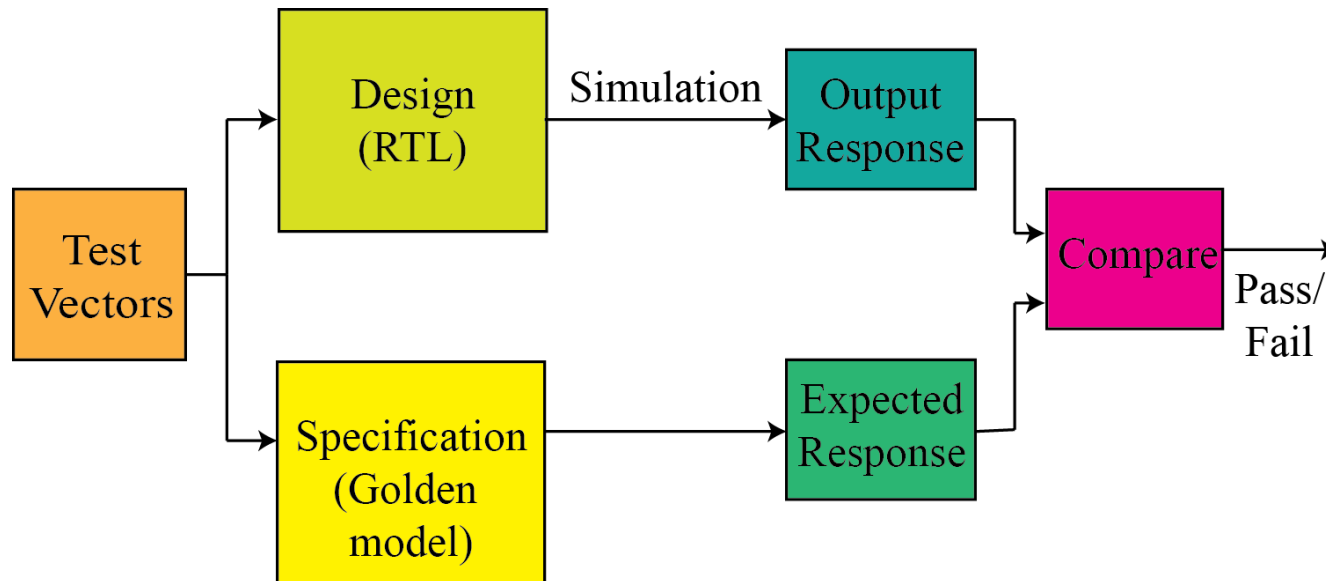
- Verification is done multiple times throughout a VLSI design flow
 - Whenever design undergoes some changes, verification must be done
 - If verification fails remedial action can be taken immediately

- Significant effort of VLSI design flow is spent on verification

Verification Techniques: Simulation

Is design (RTL) producing the same results as given in the specification?

Simulation: technique for ensuring the functional correctness of the RTL using *test vectors*



- Merits of simulation:-based verification: fast and versatile
- Demerit of simulation-base verification: incompleteness

- Test vectors: sequence of zeroes/ones and when they appear is specified at the inputs
- Obtain the response for the given RTL using a simulator
- Expected output is computed using another model (C, C++, MATLAB, etc.)
- Compare output response and expected response

Verification Techniques: Model Checking

Is design (RTL) producing the same results as given in the specification?

Model checking: technique for ensuring the functional correctness of the RTL using *formal methods*

- Formal methods: establish the proof of a given property using formal mathematical tools such as deductions
- Once we have proven a property mathematically, it is guaranteed to hold for all test stimuli.

Verification using formal methods:

- Merits: completeness
- Demerits: computationally difficult

Property Checking/Model Checking

- Define *properties* that must be satisfied for a given *specification*
- Check whether *properties* are being satisfied in the implemented design RTL using formal methods

Example:

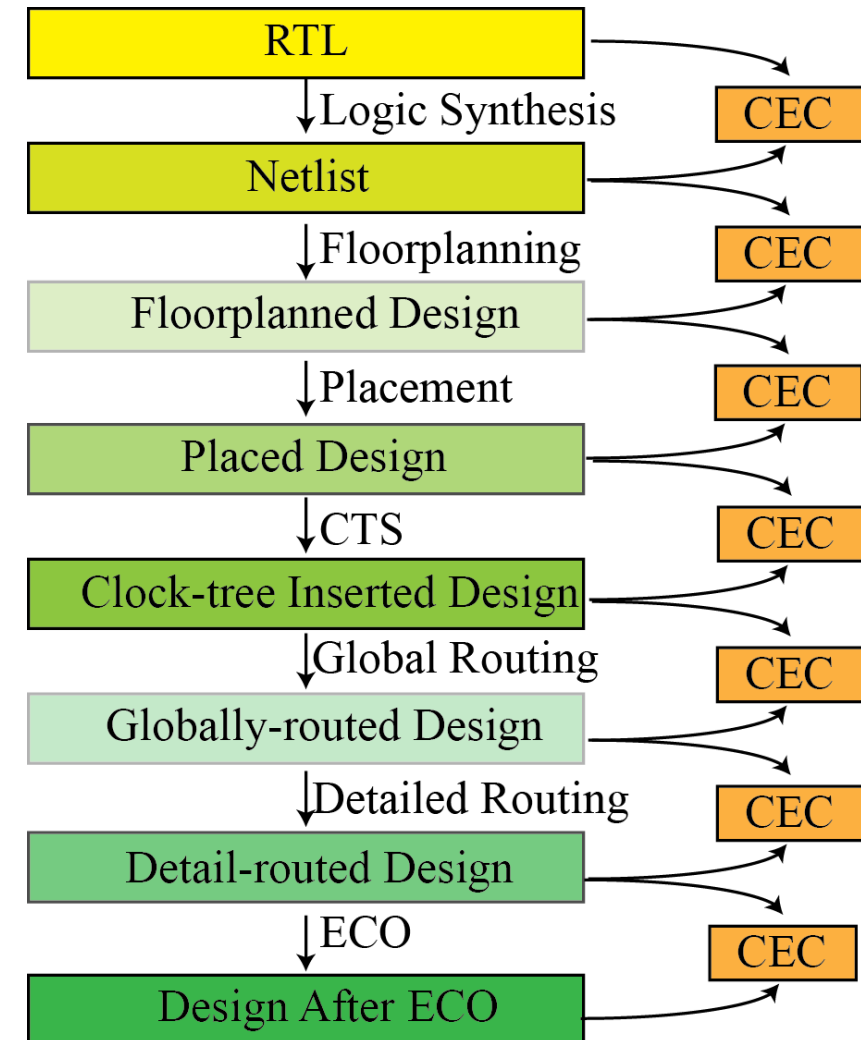
- A traffic controller should generate signals such that “*Not more than one light should be green*”
- In a resource sharing environment: “*Request for a resource should be granted eventually*”

Verification Techniques: Combinational Equivalence Checking

Will the RTL and the netlist generated by a logic synthesis tool always produce the same (equivalent) output?

Combinational equivalence checking (CEC): establishes the functional equivalence of two models using *formal methods*

- CEC required whenever non-trivial design changes occur
- Carried out multiple times in a design flow



References

- S. Saurabh, “Introduction to VLSI Design Flow”. Cambridge: *Cambridge University Press*, 2023.